EEG Artifact Simulation:
Implementation of a selection of artifacts, including eye movements, power line noise, and drifts

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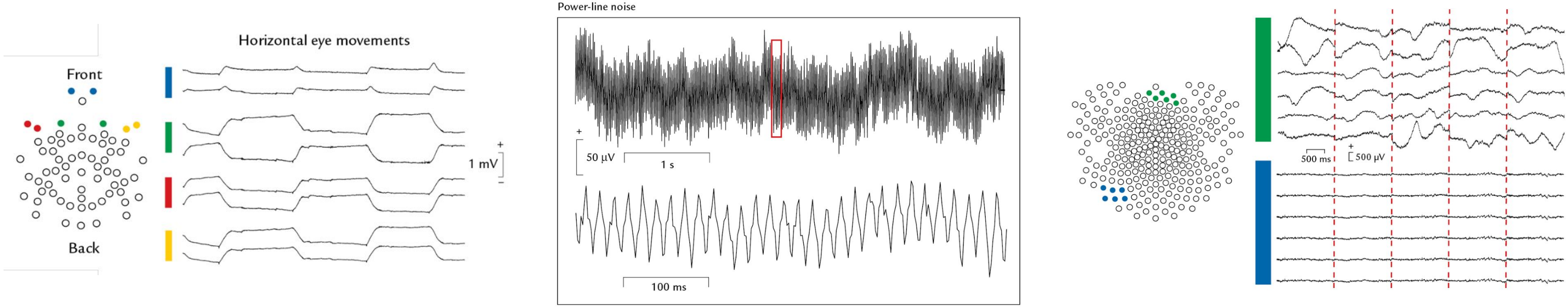
Motivation

- Artifacts - undesirable in EEG recordings



Eye movement, power line noise, and sweating artifacts. Source: Hari, R., & Puce, A. (2017)

- Simulation: improving analysis & artifact removal toolboxes
  - Specify ground truth, analyze, compare
  - Data should contain artifacts (beyond random noise), like real recordings

Related Work - Past approaches to artifact simulation

- From scratch for individual studies
- Clean(ed) or simulated data ➪ ‘extract-project-add’ artifact from real data
  - Still dependent on real data
  - Less control over specifics (e.g. “eye movement from X to Y”)
- Often in MATLAB

➪ Need for *open-source, reproducible, standardized but flexible* artifact simulation

- Research Project (03.2025 @ CCS)
  - CRD & Ensemble models, eye-position-change simulation

References: Klados and Bamidis (2016), Anzolin et al (2021), du Toit et al (2023), Downey and Ferris (2023), Kim et al (2025).

Research Questions 🕵️‍♀️

1. What are the characteristics of the measured EEG artifact potentials during an EEG recording?
2. How can we simulate these artifacts?

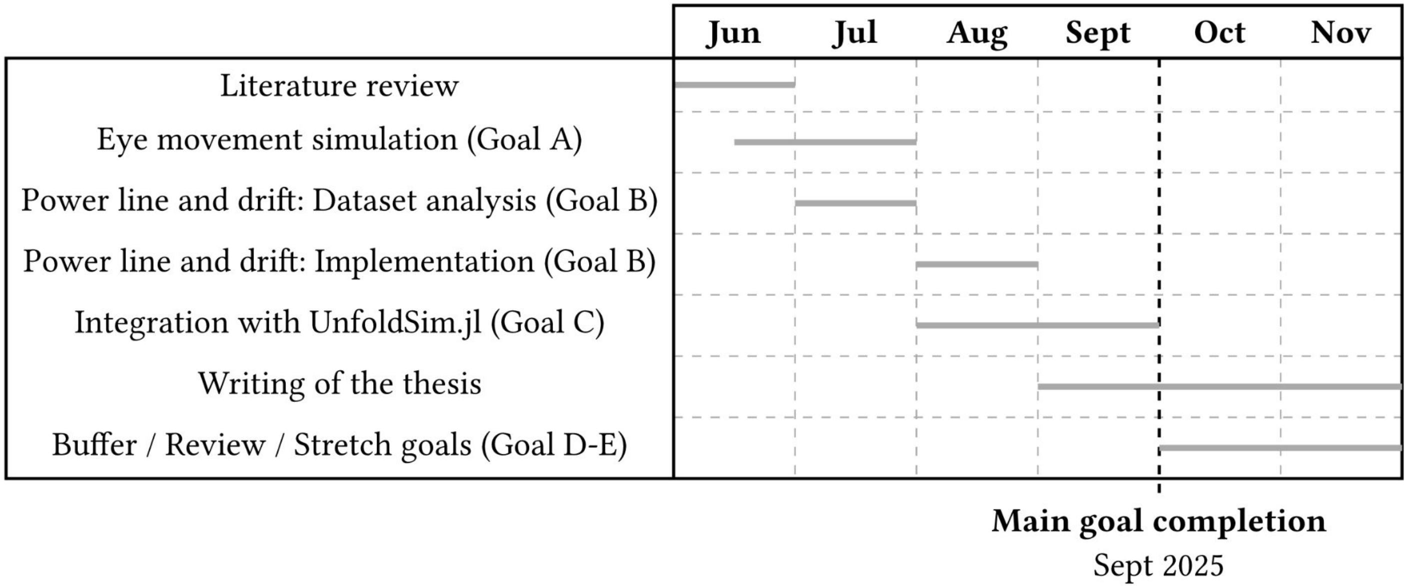
Goals & Approach

A. Eye movement trajectories: using Research Project work as base
B. Power line noise & drifts:
   a. Power line noise: simulate sine waves
   b. Drift: replicate characteristics found from literature review & data analysis
C. Code Interfaces:
   a. UnfoldSim-compatible
   b. Higher-level properties of simulation (onset, occurrence freq, ...)

Stretch goals (D,E) : Evaluation, other artifacts

Planned Schedule

Rough plan (to be adjusted as the thesis period progresses):



Recap

- EEG simulation packages: brain sources & noise but not (often) artifacts
- Current artifact simulation: non-standard, non-reproducible, closed source
- This thesis:
  - Simulate eye movement artifacts, power line noise, drifts;
  - Integrate with UnfoldSim.jl

🤔 Questions?

References

- Hari, R., & Puce, A. (2017). Meg-eeg primer. Oxford University Press, Incorporated. https://ebookcentral.proquest.com/lib/uni-stuttgart/detail.action?docID=5746005.
- Kim, H., Chang, C.-Y., Kothe, C., Iversen, J. R., & Miyakoshi, M. (2025). Juggler's ASR: Unpacking the principles of artifact subspace reconstruction for revision toward extreme MoBI. Journal of Neuroscience Methods, 420, 110465. https://doi.org/10.1016/j.jneumeth.2025.110465
- Harmening, N., Klug, M., Gramann, K., & Miklody, D. (2022). HArtMuT—modeling eye and muscle contributors in neuroelectric imaging. Journal of Neural Engineering, 19(6), 066041–066041. https://doi.org/10.1088/1741-2552/aca8ce
- Klados, M. A., & Bamidis, P. D. (2016). A semi-simulated EEG/EOG dataset for the comparison of EOG artifact rejection techniques. Data in Brief, 8, 1004–1006. https://doi.org/10.1016/j.dib.2016.06.032

References

- Anzolin, A., Toppi, J., Petti, M., Cincotti, F., & Astolfi, L. (2021). SEED-G: Simulated EEG Data Generator for Testing Connectivity Algorithms. Sensors, 21(11), 3632. https://doi.org/10.3390/s21113632
- du Toit, W., Venter, M., & Vandenheever, D. (2023). Semi-synthetic EEG Data for the Evaluation of Linear EEG Cleaning Methods. International Journal Bioautomation, 27(4), 193–214. https://doi.org/10.7546/ijba.2023.27.4.000907
- Downey, R. J., & Ferris, D. P. (2023). iCanClean Removes Motion, Muscle, Eye, and Line-Noise Artifacts from Phantom EEG. Sensors, 23(19), 8214. https://doi.org/10.3390/s23198214
- Marathe, M. (2025). Simulation of Eye Movement Artefacts in EEG: introducing the Ensemble and Corneo-Retinal Dipole methods. https://github.com/s-ccs/2024-ocular-artifacts-simulation/releases/tag/rp-report